Pasit Khumsena

0918683540 - web.pasit.kh@gmail.com - github.com/WebBacillus - leetcode.com/u/webbalaka

Computer Engineering student graduating in December 2026 with hands-on experience developing LLM-powered applications and RAG-based customer support systems. Strong backend engineering foundation with experience in AI workflow design, document processing, and API integrations. Seeking an entry-level AI Engineer role to build reliable, scalable AI solutions.

EDUCATION

Chulalongkorn University Expected: December 2026

B.Eng. in Computer Engineering (GPAX: 3.71 / 4.0)

Relevant Coursework: Software Engineering for ML Systems, Digital & AI Transformation, Introduction to Data Science and Data Engineering, Data Structures and Algorithms, System Design for Software Development, Software Engineering,

EXPERIENCE

True Corporation
August 2026 – Present

AI Engineer (Intern)

- **Product Ownership:** Drive product design and project planning for an AI customer support agent; define project scope, development timelines, and customer workflows, supported by market and competitor research.
- **Thai PDF Processing & Unicode Recovery:** Develop a document processing pipeline using Docling and font-aware Unicode recovery to resolve unknown Thai characters and prepare device manuals for a RAG knowledge base.
- **AI Customer Support Agent:** Develop a RAG prototype for TrueX smart home support using LLMs and Qdrant; implement device retrieval and evidence validation, and design guided troubleshooting workflows with escalation to human support.

National Interbank Transaction Management and Exchange (NITMX)
April 2025 – July 2026

Software Engineer (Part-time)

- **Legacy Code Modernization & Security:** Successfully upgraded critical legacy applications from Java 11 to Java 25, refactoring the codebase to leverage modern language features while identifying and mitigating security vulnerabilities.
- **Anti-Mule Account:** Collaborated on a national-level project detecting mule accounts; designed and executed comprehensive automated API testing utilizing Postman and Bruno to validate complex input fields and verify XML outputs for data integrity.
- **Interbank Financial Adjustment Web Application:** Developed a secure interbank financial adjustment web application using Go, Clean Architecture, and Hexagonal Architecture.
- **Blockchain Infrastructure:** Contributed to the design of a blockchain network using Hyperledger Besu; developed full-stack features with NestJS and Next.js, and established a resilient Disaster Recovery (DR) plan by provisioning backup nodes and automating deployment using Ansible playbooks.

PROJECTS

- **Scalable ML Classification Pipeline:** Architected a robust MLOps pipeline for scalable image classification using PyTorch, Airflow, and FastAPI; implemented automated model training, versioning, and inference workflows with ResNet34 and EfficientNetV2-S to ensure system reliability and high-throughput production readiness.
- **Scopus Gender Prediction:** Scraped and cleaned data from Scopus and Crossref into sparse matrices, trained a Random Forest Classifier for gender prediction achieving an F1 score of 0.87, and built interactive dashboards with Streamlit and Plotly to visualize insights.
- **NECTEC Medical AI Research:** Constructed a multi-label chest X-ray diagnostic system; addressed significant dataset imbalances through advanced data augmentation and balancing techniques, selecting EfficientNetV2 to ensure superior clinical accuracy.
- **SCB Tech X Loan Origination:** Engineered high-performance Go microservices and Temporal workflows to automate manual loan processes; integrated Azure Blob Storage and PostgreSQL to enhance data security and operational efficiency.

TECHNICAL SKILLS

Languages: Go, Python, C++, TypeScript, JavaScript, Java

Databases: PostgreSQL, MongoDB, MySQL

Backend & Architecture: REST API, gRPC, Microservices, Clean/Hexagonal Architecture, Kafka, RabbitMQ

AI & ML: PyTorch, Torchvision, Scikit-learn, Pandas, NumPy, Apache Airflow, MLflow, Streamlit, Plotly

Generative AI: OpenAI API, LLM Tool Calling, RAG, Text-to-Speech (TTS), n8n, LangChain, Langraph, Qdrant

Testing & Quality: Unit, Integration, and Load Testing, promptfoo (Red Teaming/Agent Evaluation)